

Autodesk® Scaleform®

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Autodesk® Scaleform® 4.4

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1.1	AMP 	1
1.2	AMP 	2
2	AMP ~	10
2.1		10
2.2		11
2.3		12
2.4	$\dot{p}$	13
2.5		19
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3	∇	21
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5.1	Debug	23
5.2	L	23
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5.4		24
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7		26

1

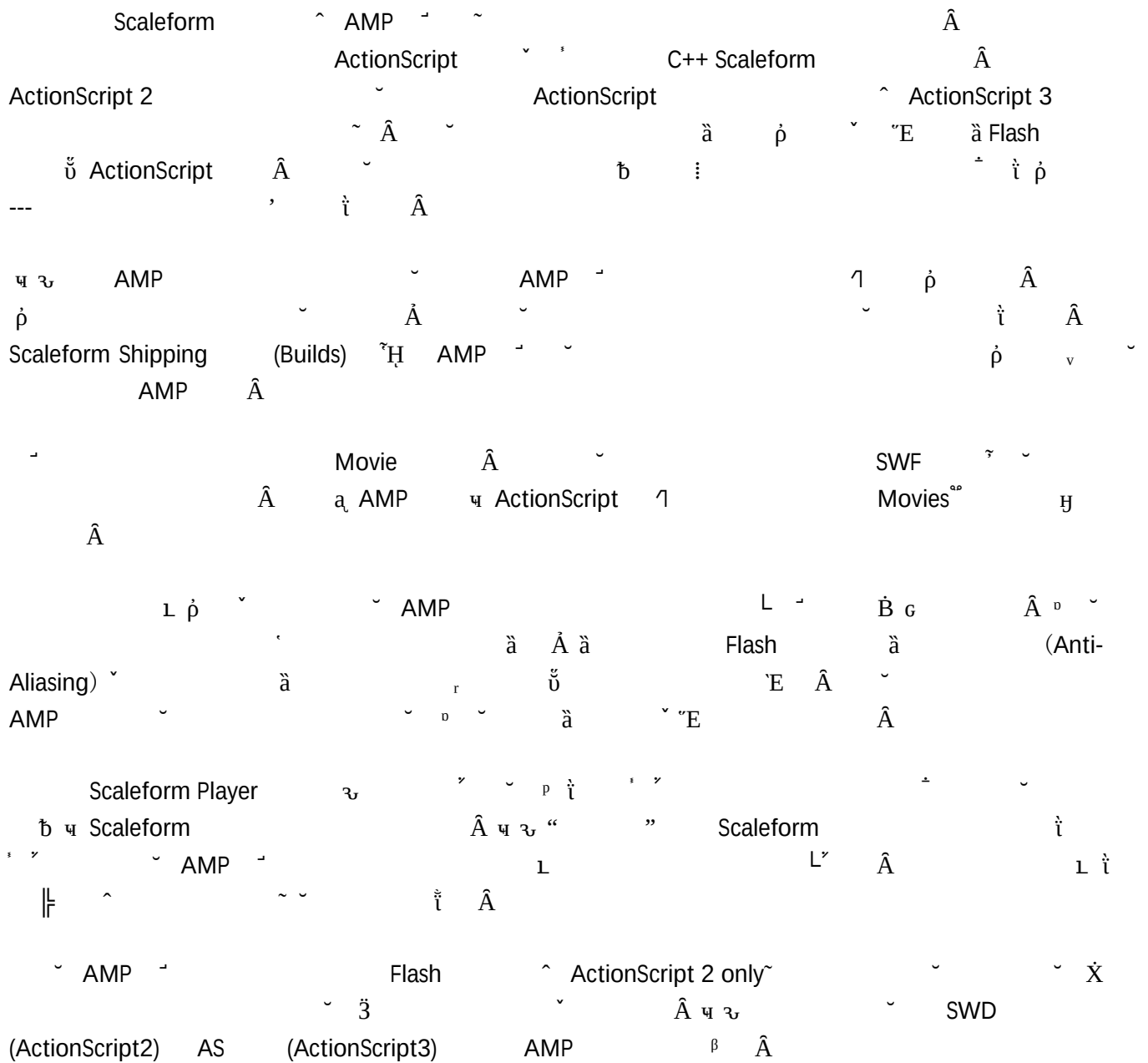
AMP™ Scaleform® E CPU a ' à Scaleform
p ' Â L p v ' H ~
v Â AMP ì ActionScript v ^ G Â

1' Scaleform AMP

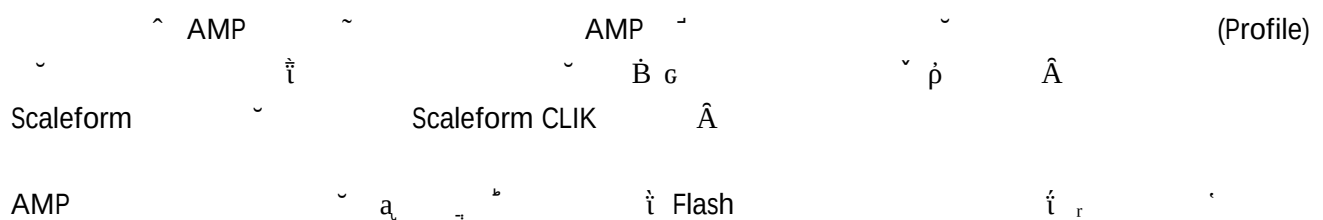
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1.1 AMP ~



1.2 AMP



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Scaleform $\hat{A}$

0 ‘ X “ ” (Anti-alias for readability)

$$0 \leq L(H) \leq H, \quad \hat{A} = H - L(H).$$
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“(OK)

XML

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“-adviser <AdviserFilePath>”

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o (Clear) || A ~ 4 AMP p
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o A (Pause) A p v A
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o (Next Frame) 1 || ,
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o (Last Frame) || A
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x “ ”(Profiling) t ‘ ”H G A

o (Detailed Profiling) ^ p ~ ‘ ~ AMP
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a “ ”(Medium) v “ ”(High) A y a AMP C++ A
ActionScript v A 3 y C++
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“ ”(Code) A , ~ (SWD) A
ActionScript 3 , A

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v A (Zooming out) ”H
~ (Zooming In) ”H + A 3 a
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x “ ”(Status) ‘
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UI
“ UI”(Reset UI)

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1 "H à à à a à à "H à
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 G H Â Â (Best Practices

Guide)

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“ ” (Render Summary)

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- MeshBatch[^] ~ ' “ ” √ || 1
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- Context[^] ~ ' 1 ρ ~ 9 Â
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- NodeData[^] ~ ' g ^ ||
~ ρ Â
- TreeCache[^] ~ ' ^ ~
- TextureManager[^] ~ ' "E ρ Â "E
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- Text[^] ~ ' ^ ã ~ ã html ã
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- CharDefs[^] H ~ ' g H ~ ρ Â
- ShapeData[^] ~ ' g H ρ Â
- Tags[^] ~ ' g ActionScript ρ
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- ASBinaryData[^] ActionScript H L ~ ' ActionScript || 1 1 ρ Â
- MD_Other[^] ì H ~ ' ì MovieDef Â
- o MovieView[^] ~ ' ù ù ActionScript ρ Â
- MovieClip[^] ◀ ~ ' g ù ρ Â
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x ASString[^] ActionScript ~ ' ActionScript 1 ρ Â
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- XML[^] g XML ρ Â
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x Vtable^ ~ ' ActionScript 1 ρ Â
x SlotInfo^ ~ ' G ActionScript " ρ Â
ClassTraits ~ InstanceTraits SlotInfo ~ B G " " (Class)
" " (Instance) ρ Â ~ ClassTraits ~ InstanceTraits
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o SlotInfoHash^ ف ~ ' + ActionScript ف
(Hash Table) a ρ Â
x ClassTraits^ ~ ' ActionScript ρ ρ Â
x Class^ ~ ' ActionScript " ρ Â
x InstanceTraits^ " ~ ' ActionScript ρ
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x Instance^ " ~ ' ActionScript " ρ Â
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“ ” (Images)

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A Scaleform 1 E v E
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“ ”(Fonts)

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1. AMP Gfx::System Gfx::System::Init Â ~ X
 Saleform::System Scaleform~
 Gfx::AMP::Server::Init() Gfx::AMP::Server::Uninit() ~ AMP Â
2. AMP || Windows à Xbox360 à PS3 à Linux à Mac, Wii, Android iPhone/iPad, 3DS,
 PS Vita, WiiU Â AMP ~ ~ 1 Â ~ ~
 AMP ~ IP ~ ~ ~ IP Â
- x g Xbox360 ~ IP ~ IP Â IP Xbox
 360 ~ ~ Â
- x g PS3 ~ ~ IP ~ ~ IP ~ g
 IP Â
- x g Android ~ ~ ~ ~ ~
 AndroidManifest.xml
 <uses-permission android:name="android.permission.INTERNET" />
- x h PC ^ WiFi ~ WiiU Â h PC ~ 6003
 Â ~ ~ TCP/IP h Â
- x Wii a AMP ~ a p NDEV USB ^ COM ~
 L ~ PC Â IP ~ "wii" ~
 Wii AMP ~ Â ~ RVL_SDK/X86/bin Â ~
 AMP hio2.dll Â
3. ~ ^ Unreal Engine 3~ ~ ~ a AMP ~
 Flash p AS2 SWD Â ~ ~ SWD AMP
 b ~ ~ AMP ~ SWF ~ ~ ~ Â

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1. AMP Scaleform AMP
GfXConfig.h SF_AMP_SERVER Scaleform AMP Scaleform

Ptr<GFx::AMP::ServerState> serverState = *SF_NEW GFx::AMP::ServerState();
serverState->StateFlags &= GFx::AMP::Amp_Disabled;
AmpServer::GetInstance().UpdateState(serverState); AMP
2. AMP (Network Socket) AMP
Ws2_32.lib Windows Xnet.lib Xbox360 libnetctl PS3
Linux . AMP
3. AMP AMP AMP
AMP AMP AMP
AMP::Server::GetInstance().SetInitSocketLib(false) AMP
AMP AMP AMP
4. AMP AMP AMP Scaleform
AMP AMP::Server::GetInstance().SetHeapLimit()
AMP
5. AMP 7534 AMP
AMP::Server::GetInstance().SetListeningPort() AMP
6. AMP AMP AMP
AMP::Server::GetInstance().SetConnectionWaitTime AMP


```
caps.SetToggleBatch(true);
caps.SetToggleStrokeType(true);
caps.SetTogglePause(true);
caps.SetToggleWireframe(true);
AMP::Server::GetInstance().SetAppControlCaps(&caps);
```

5.3

AMP

AMP::Server::GetInstance().SetConnectedApp

```

    AMP
    AMP
    AMP
    AMP::Server::GetInstance().SetState
        o Amp_RenderOverdraw
        o Amp_App_Wireframe
        o Amp_App_Paused
        o Amp_App_FastForward
    AMP::Server::GetInstance().SetAaMode
    AMP::Server::GetInstance().SetStrokeType
    AMP::Server::GetInstance().SetCurrentLocale
    AMP::Server::GetInstance().SetCurveTolerance
```

6 AMP

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AMP GFX_AMP_SERVER GfX
AMP Platform::RenderThread::drawFrame
AMP AmpServer::GetInstance().AdvanceFrame

2. " G· AmpClient bm644 !{I @Ç4ž8.è 4 !{I @Ç4žI 0ŧb®F &g2&ií0ÇQ¢3ñ í4 !{&`F?
AmpServer::AdvanceFrame() U@_È 8...hf í

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3. Windows/PS3/Xbox AMP

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Platform::RenderThread::drawFrame

AMP

AmpServer::GetInstance().AdvanceFrame

4. AMP É!ä)·9f8?8...9 "ë •Lag.4¥9fh 8...hf (AMP::Server::WaitForAmpConnection) ũ

AMP

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AMP (profile level) 'i'
1MB GfX::AMP::Server::SetHeapLimit

5.

AMP 1MB
 AmpServer::SetHeapLimit AMP
 Gfx

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- x [AMP a \(AMP Use Cases\)](#) Â
- x [Scaleform Reference](#) – / Â
- x [AMP Forum](#) – g 1 v Â